SHAKTI RAIL COMPONENTS PVT. LTD.

MATERIAL SPECIFICATION — STRUCTURAL STEEL S355 (BOGIE SIDE FRAMES, BOLSTERS)

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1. SCOPE. This specification sets the acceptance limits for steel grades S355J0 and S355J2 (EN 10025-2) used for bogie side frames, bolsters and welded brackets on drawings SRC-BF-2041, SRC-AB-1107 and derivatives. Limits apply to the ladle analysis and to mechanical tests reported on the mill test certificate (EN 10204 type 3.1).
2. CHEMICAL COMPOSITION — ladle analysis, maximum % by mass (clause 2.1, Table 1).

Grade / thickness t	C	Si	Mn	P	S	Cu	CEV	N
S355J2, t <= 30 mm	0.20	0.55	1.60	0.025	0.025	0.55	0.45	-
S355J2, 30 < t <= 100	0.22	0.55	1.60	0.025	0.025	0.55	0.47	-
S355J0, t <= 30 mm	0.20	0.55	1.60	0.030	0.030	0.55	0.45	0.012
S355J0, 30 < t <= 100	0.22	0.55	1.60	0.030	0.030	0.55	0.47	0.012

CEV = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15. Where the certificate does not state the section thickness, the t <= 30 mm row applies.

3. MECHANICAL PROPERTIES — tensile test, longitudinal (clause 3.1, Table 2).

Thickness t	Yield ReH min (MPa)	Tensile Rm (MPa)	Elongation A min (%)
t <= 16 mm	355	470 - 630	22
16 < t <= 40 mm	345	470 - 630	22
40 < t <= 63 mm	335	470 - 630	21

4. IMPACT TEST — Charpy V-notch, average of three specimens (clause 3.2, Table 3).

Grade	Test temperature	Minimum energy KV
S355J2	-20 C	27 J
S355J0	0 C	27 J

An impact result at a higher temperature than specified does not demonstrate conformity (e.g. 0 C results for S355J2).

5. ACCEPTANCE. Every heat must be covered by a type 3.1 certificate. Any value outside these limits, or a required test not reported, makes the heat non-conforming: quarantine the material and raise an NCR (see SRC-QP-07). Concessions require the QA Head's written approval.

Approved: N. Verma (QA Manager) Date: 14 Aug 2026